

ERIK PAK

pak.erik@icloud.com | Chicago, IL | (718) 644-8343 | | <https://www.linkedin.com/in/erikpak>

OVERVIEW: I am a Data Scientist committed to creating impactful business solutions through advanced analytics. I specialize in data acquisition, processing, and model development, using Python, R, and SQL to convert intricate data into actionable insights that drive measurable results. With a strong passion for innovation, I am dedicated to propelling organizational growth and success, which fuels my passion and underscores my long-term commitment.

SKILLS & EXPERTISE:

- Data Science Algorithms
- Databases
- Data Wrangling
- Cloud & Big Data
- Machine Learning
- Data Privacy Protection
- Feature Engineering
- Visualization
- AI/ML
- ETL Pipelines
- Statistical Analysis
- Project Planning

SKILLS:

Languages: Python, SQL, R, HTML
Developer Tools: Jupyter Notebook, Git, Visual Studio Code, Docker
Data Collection: JSON, CSV, API, Web-Scraping, AWS, Hadoop (Map/Reduce), PySpark
Database Management: MS SQL Server, PostgreSQL, MongoDB, MySQL, Neo4j
Machine Learning Framework: Scikit-learn, TensorFlow, PyTorch
Big Data & Cloud: AWS, Hadoop (Map/Reduce), PySpark
Machine Learning: Classification, Clustering, Boosting, CART, Ensemble Model, NLP, KNN, PCA, LDA (**Linear Discriminant Analysis**), One-Class SVM, SVD (Singular Value Decomposition), FAMD (Factor Analysis of Mixed Data), MCA (Multiple Correspondence Analysis), Gaussian Mixture
Imputing Tools: SimpleImputer, IterativeImputer, KNN imputer, Target Encoding with K-fold Cross-Validation
Discretization Tools: Gaussian Mixture (AIC & BIC), KBinsDiscretizer
Data Visualization: Tableau, Matplotlib, Seaborn, ggplot2, MS Reporting Service
Libraries: Scikit-learn, TensorFlow, OpenCV, PILLOW, NumPy, Pandas, matplotlib, seaborn, TfidfVectorizer, CountVectorizer, SciPy, tidyverse, MASS, ggplot2, car, leaps, glmnet, ca, dplyr, QuantPsyc, Bayesian Optimization (skopt), NLP

TRAINING: **DePAUL UNIVERSITY** 2022 to 2024
Data Science Projects

HR Analytics - Employee Attrition & Performance

- Based on quantitative and qualitative data, created and analyzed predictive models used to identify potential employee outcomes. Techniques used to determine model accuracy included Principal Factor Analysis, Parallel Analysis, Correspondence Analysis, Linear Regression (Lasso, Ridge, and Elastic Net), Linear Discriminant Analysis,

Hand Gesture Audio Device Control

 - Using webcam input, CNN system continuously monitors hand gestures and triggers

Movie Recommender System

 - Integrating regression analysis for movie revenue and classification algorithms for movie (SVD) based on user's previous movie ratings.

Lane and Curve Detection

 - Augmented autonomous driving system using real-time image segmentation and model optimization using Gaussian, Sobel, Prewitt, and Laplacian filters and histogram Equalization.

EXPERIENCE: **DePAUL UNIVERSITY** 2024 to Present
Graduate Research Assistant

- Graduate research project with professor utilizing UNOS (Thoracic) transplant data with a focus on improving post-transplant aftercare efficiency and long-term patient outcomes. Applied advanced problem-solving, data analysis, and data wrangling techniques to uncover actionable insights. Leveraged statistical methods and domain expertise while continuously enhancing skills in medical data analytics and patient care optimization.

SPRINGBOARD 2020 to 2023
Data Science Fellow

- As a team member, designed multiple machine learning algorithms for fraud detection
- Used Kaggle API for data collection and NLP for sentiment analysis.
- Developed end-to-end pipelines in Python for streamlined data ingestion, preprocessing,

OTHER EXPERIENCE:

Prior to my professional pivot, held technology lead roles at Pfizer Pharmaceutical and Prudential Insurance, managing critical HR data systems. As an entrepreneur, managed all aspects of a customer-focused business, demonstrating strong problem-solving, communication, and relationship management skills.

- **Database Management & Data Modeling:** Database Management & Data Modeling: Extensive experience with MS SQL Server, including optimization, performance tuning, and monitoring. A strong foundation in designing and managing complex databases is crucial for building effective data models. Proficient in SQL querying, essential for efficient data handling and ETL processes. Developed internal SQL Server monitoring systems, skilled in data tracking and reporting.
- **Data Architecture & Integration:** Designed and optimized database and process architectures for business systems, creating efficient and resilient data solutions for transactional and reporting. Built data pipelines that enable seamless integration across diverse systems, enhancing data accessibility, improving data quality, and supporting analytics, including compliance with data privacy.
- **Data Quality & Assurance:** Led Payroll QA efforts for data migrations and system implementations, with a strong commitment to data quality and reliability. Implemented data validation strategies to ensure seamless transitions and maintain high data integrity standards.
- **Business Intelligence & Analytics:** Managed HCM and internal HR databases (EEO, succession planning, performance), extracting actionable insights from organizational data to drive data-driven decisions. Expertise in generating reports and descriptive statistics to support organizational improvements.
- **Project Management & Collaboration:** Led cross-functional teams in complex data-related payroll projects, employing robust project management methodologies to ensure on-time delivery while maintaining strict data security and confidentiality. Enhanced team collaboration and communication to guarantee accurate data migration and compliance with privacy regulations.
- **Stakeholder Collaboration:** Cultivated strong relationships with clients, vendors, and internal departments to drive the implementation of data-driven solutions. Effectively managed stakeholder expectations and communication to ensure alignment and success in infrastructure projects.
- **Process Optimization:** Leveraged strong analytical and problem-solving skills to improve ETL performance between SAP & PeopleSoft, streamlining HR data processing. Resolved technical bottlenecks, leading to a 10% reduction in weekly batch processing time and enhanced data quality for critical HR functions.
- **Adaptability & Innovation:** Proactively learned and implemented new features and functionalities in MS SQL Server and MS Reporting Services to optimize the performance and monitoring of critical databases. Provided data-driven insights, enhancing accessibility and enabling informed decision-making. Demonstrated a commitment to staying ahead of industry advancements in data management.

EDUCATION:

DePaul University - M.S. Data Science

Fairleigh Dickinson University - M.B.A. - Human Resources

Newark College of Engineering - B.S. Engineering